

Sine rule



1

a No

b Yes

c Yes

d No

2

a $\frac{a}{\sin A} = \frac{b}{\sin B}$

b $\frac{p}{\sin P} = \frac{q}{\sin Q} = \frac{r}{\sin R}$

3

a $\sin A = \frac{x}{b}$

b $x = a \sin B$

c $\frac{\sin A}{a} = \frac{\sin B}{b}$

Unknown sides

4

a $a = 10.50$

b $a = 6.85$

c $a = 3.02 \text{ m}$

d $a = 711.73 \text{ cm}$

e $a = 11.29 \text{ cm}$

5

a $x = 13.1$

b $x = 12.9$

6

a i $a = 16.66$

ii $a = 16.66$

b i $a = 17.14$

ii $a = 17.14$

c i $a = 30.09$

ii $a = 30.09$

7

a $HK = 733.17 \text{ km}$

b $KJ = 444.55 \text{ km}$

8

a $\angle A = 75.78$

b $b = 3.152 \text{ m}$

c $c = 5.723 \text{ m}$

Unknown angles

9

a $x = 26.0^\circ$

b $x = 32.7^\circ$

c $x = 50.7$

d $x = 30.4^\circ$

e $x = 20.4^\circ$

f $x = 30.3^\circ$

g $x = 43.2^\circ$

h $x = 149.1^\circ$

i $x = 133.13^\circ$

j $x = 23.5^\circ$



10

a $B = 74^\circ$

b $B = 31^\circ$

11

a $x = 19.54^\circ$

b 48.46°

12

a $x = 41$

b 139

13 $\theta = 33.84^\circ$

Applications

14

a $\angle OBA = 39^\circ$

b $k = 21.76$

15 $x = 53.6$ m

16

a $\angle A = 38^\circ$

b $c = 22.71$ m

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d 1.9 min